Neural SDE: Stabilizing Neural ODE Networks with Stochastic Noise

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Motivation

Many physical systems, such as pendulums, fluid flows, and robots, can be described by differential equations:

$$\frac{dx(t)}{dt} = f(x(t), t)$$

where $x(t)$ is the system state and f represents the governing dynamics.

Main Idea of Neural ODEs

Instead of explicitly knowing f, we approximate the dynamics using a neural network f_θ.

Problem of Neural ODEs

Neural ODEs are deterministic, making it difficult to incorporate stochastic regularization. Small input perturbations may also be amplified.

Introduction of Neural SDEs

Add a stochastic term to the Neural ODE:

$$\boxed{dx_t = f_\theta(x_t, t)\, dt + G_\phi(x_t, t)\, dB_t}$$

Study how stochastic noise affects **generalization, robustness, and stability** against input perturbations.

From ResNet to Neural ODE

1. Residual Neural Network (ResNet)
A ResNet updates hidden features by adding a learned correction to the current state through a skip connection:

$$h_{n+1} = h_n + f(h_n; w_n),$$

2. Continuous-depth interpretation
Interpret the layer index as a discrete depth variable, $t_n = n\Delta t$, with $h_n \approx h(t_n)$.
Writing the residual update with a depth step Δt, gives the forward-Euler discretization:

$$h_{n+1} = h_n + \Delta t\, f(h_n, t_n; w)$$

In the continuous-depth limit, $\Delta t \to 0$:

$$\frac{\mathrm{d}h(t)}{\mathrm{d}t} = f(h(t), t; w)$$

Advantages of Neural ODEs: Neural ODEs can reduce trainable parameters by using the same network across continuous depth instead of separate networks for each layer, and improve numerical accuracy by automatically adjusting the solver step size.

3. Integral representation Integrating the dynamics from depth s to t:

$$h_t = h_s + \int_s^t f(h_\tau, \tau; w)\ \mathrm{d}\tau$$

The hidden features evolve continuously, with their dynamics governed by a neural-network-parameterized vector field f.

Why Introduce Stochasticity into Neural ODEs?

1. Problem with Neural ODEs

Neural ODEs describe deterministic hidden-state dynamics:

$$dh_t = f(h_t, t; w)\, dt.$$

However, a small perturbation in the initial state may grow as it propagates through network depth:

$$h_0^e = h_0 + \varepsilon_0, \qquad \varepsilon_t = h_t^e - h_t.$$

$$\|\varepsilon_0\| \ll 1 \quad \not\Rightarrow \quad \|\varepsilon_T\| \ll 1.$$

Therefore, Neural ODEs are not necessarily robust to input perturbations.

2. Motivation from Stochastic Regularization

In discrete neural networks, methods such as dropout and Gaussian noise injection are commonly used to reduce overfitting and improve generalization.

However, these stochastic regularization methods cannot be directly applied to deterministic Neural ODEs.

3. Proposed Solution: Neural SDE

To address these limitations, the paper introduces a stochastic term into the continuous dynamics:

$$dh_t = \underbrace{f(h_t, t; w)\, dt}_{\text{Drift}} + \underbrace{G(h_t, t; v)\, dB_t}_{\text{Diffusion}}$$

The main idea is to use stochastic noise to improve generalization and control the effect of input perturbations.

1-D Toy Example: How Can Noise Stabilize an ODE?

Consider the scalar SDE:

$$\boxed{dx_t = x_t\,dt + \sigma x_t\,dB_t, \quad x_0 \neq 0.}$$

1. Without noise: $\sigma = 0$

$$dx_t = x_t\,dt \quad \Longrightarrow \quad x_t = x_0 e^t.$$

Since $|x_t| \to \infty$, the zero equilibrium is unstable.

2. With multiplicative noise: $\sigma > 0$

Apply Itô's formula to $\log|x_t|$:

$$d\log|x_t| = \frac{dx_t}{x_t} - \frac{(dx_t)^2}{2x_t^2}.$$

Since $(dx_t)^2 = \sigma^2 x_t^2\,dt$:

$$d\log|x_t| = \left(1 - \frac{\sigma^2}{2}\right)dt + \sigma\,dB_t.$$

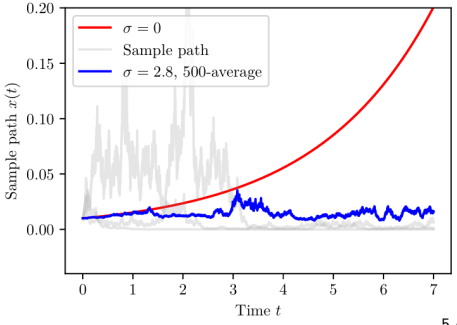
Integrating gives

$$x_t = x_0 \exp\left[\left(1 - \frac{\sigma^2}{2}\right)t + \sigma B_t\right]$$

3. Stability condition

$$\frac{B_t}{t} \xrightarrow{\text{a.s.}} 0 \quad \Longrightarrow \quad \lim_{t\to\infty} \frac{\log|x_t|}{t} = 1 - \frac{\sigma^2}{2}$$

$$\boxed{\sigma^2 > 2 \quad \Longrightarrow \quad x_t \to 0 \text{ exponentially a.s.}}$$



Neural SDE: Mathematical Formulation

$$dh_t = f(h_t, t; w)\, dt + G(h_t, t; v)\, dB_t$$

Components of the Neural SDE

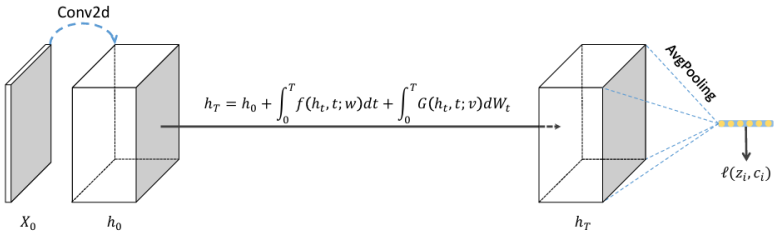
- $h_t \in \mathbb{R}^n$: hidden-state vector evolving continuously with network depth t.
- $f(h_t, t; w) \in \mathbb{R}^n$: **drift function**, parameterized by neural-network weights w, describing deterministic state evolution.
- $B_t \in \mathbb{R}^m$: **Brownian motion**, introducing continuous-time randomness.
- $G(h_t, t; v) \in \mathbb{R}^{n \times m}$: **diffusion matrix**, controlling the magnitude, direction, and state dependence of injected noise.

Integral representation:

$$h_T = h_0 + \underbrace{\int_0^T f(h_t, t; w)\, dt}_{\text{Drift contribution}} + \underbrace{\int_0^T G(h_t, t; v)\, dB_t}_{\text{Noise contribution}}.$$

Key point: Setting $G = 0$ recovers the deterministic Neural ODE.

Neural SDE: Architecture and Forward Propagation



1. Apply a convolutional layer to the input image x to obtain the initial hidden state.
2. Integrate the Neural SDE from depth 0 to T (Numerically using Euler–Maruyama).
3. Apply average pooling followed by a linear classifier to obtain the prediction.

Neural SDE: Discretization and Noise Injection

1. Continuous-Time Neural SDE

The hidden state evolves through deterministic dynamics and stochastic noise:

$$dh_t = \underbrace{f(h_t, t; w)\, dt}_{\text{Deterministic}} + \underbrace{G(h_t, t; v)\, dB_t}_{\text{Stochastic}}$$

2. Discretization Using Euler–Maruyama

Similar to Euler integration for ODEs, Euler–Maruyama approximates the SDE over a time step Δt:

$$h_{k+1} = h_k + f(h_k, t_k; w)\Delta t + G(h_k, t_k; v)\Delta B_k.$$

Since Brownian increments satisfy

$$\Delta B_k \sim \mathcal{N}(0, \Delta t I_m), \qquad \Delta B_k = \sqrt{\Delta t}\, z_k, \quad z_k \sim \mathcal{N}(0, I_m),$$

we obtain

$$\boxed{h_{k+1} = h_k + \underbrace{f(h_k, t_k; w)\Delta t}_{\text{Learned residual}} + \underbrace{G(h_k, t_k; v)\sqrt{\Delta t}\, z_k}_{\text{Random noise}}}$$

3. Connection to ResNet

- The first two terms resemble a ResNet residual update.
- The additional term injects Gaussian noise at each integration step.
- Setting $G = 0$ recovers the Neural ODE update.

Designing the Diffusion Term: Noise Models

Goal: Design $G(h_t, t; v)$ to represent different noise-injection mechanisms in continuous-depth networks.

$$\mathrm{d}h_t = f(h_t, t; w)\ \mathrm{d}t + G(h_t, t; v)\ \mathrm{d}B_t$$

1. Additive Gaussian Noise: State-independent diffusion

$$G(h_t, t) = \Sigma(t) \quad \Longrightarrow \quad \mathrm{d}h_t = f(h_t, t; w)\,\mathrm{d}t + \Sigma(t)\,\mathrm{d}B_t.$$

2. Multiplicative Noise: State-dependent diffusion

$$G(h_t, t) = \Sigma(h_t, t) \quad \Longrightarrow \quad \mathrm{d}h_t = f(h_t, t; w)\,\mathrm{d}t + \Sigma(h_t, t)\,\mathrm{d}B_t.$$

For scalar Brownian motion, $G(h_t, t) = \sigma h_t$:

$$\mathrm{d}h_t = f(h_t, t; w)\,\mathrm{d}t + \sigma h_t\,\mathrm{d}B_t.$$

3. Dropout-Style Noise: Drift-dependent diffusion

$$G(h_t, t; v)\ \mathrm{d}B_t = \sqrt{\frac{1-p}{p}}\, f(h_t, t; w) \odot \mathrm{d}B_t \quad \Longrightarrow \quad \mathrm{d}h_t = f(h_t, t; w)\ \mathrm{d}t + \sqrt{\frac{1-p}{p}}\, f(h_t, t; w) \odot \mathrm{d}B_t$$

Key point: The structure of G determines how randomness enters the hidden-state dynamics.

From Discrete Dropout to Neural SDE

1. Dropout in a Discrete ResNet Dropout randomly keeps or removes components of the learned residual:

$$h_{n+1} = h_n + f(h_n; w_n) \odot \frac{\gamma_n}{p}$$

where p is the probability of keeping a component, $\odot$ denotes elementwise multiplication, and

$$\gamma_{n,i} \sim \text{Bernoulli}(p), \qquad \gamma_{n,i} = \begin{cases} 1 & \text{with probability } p, \\ 0 & \text{with probability } 1-p. \end{cases}$$

2. Separate the Residual and Random Noise Rewrite the dropout update as

$$h_{n+1} = h_n + f(h_n; w_n) + f(h_n; w_n) \odot \left(\frac{\gamma_n}{p} - I \right).$$

The normalized dropout perturbation satisfies

$$\mathbb{E}\left[\frac{\gamma_n}{p} - I \right] = 0, \qquad \text{Var}\left(\frac{\gamma_n}{p} - I \right) = \frac{1-p}{p}.$$

3. Approximate Dropout Noise by Gaussian Noise Since $\Delta B_k = \sqrt{\Delta t}\, z_k$, $z_k \sim \mathcal{N}(0, I)$, this motivates a continuous-time SDE.

$$\frac{\gamma_n}{p} - I \approx \sqrt{\frac{1-p}{p}}\, z_n, \qquad z_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1).$$

This motivates the stochastic differential equation:

$$dh_t = f(h_t, t; w)\, dt + \sqrt{\frac{1-p}{p}}\, f(h_t, t; w) \odot dB_t$$

From Neural ODE Training to Neural SDE Training

The Training Problem: Both Neural ODEs and Neural SDEs require computing the gradient of the loss with respect to network parameters:

$$\nabla_w L(w).$$

Backpropagation Through a Numerical Solver

First, solve the dynamics forward:

$$h_0 \to h_1 \to \cdots \to h_N.$$

Then, use the chain rule backward through all solver steps $\implies$ Memory: $\mathcal{O}(N)$

Neural ODE Adjoint Method (Background)

Instead, define the adjoint as the sensitivity of the terminal loss to the hidden state: $\implies$ $a_t = \nabla_{h_t} L$

Step 1: Solve the Neural ODE forward from 0 to T to obtain h_T

Step 2: Initialize the adjoint at the final time: $\implies$ $a_T = \nabla_{h_T} \ell(h_T)$

Step 3: Solve the adjoint ODE backward from T to 0:

$$\frac{da_t}{dt} = -\left(\frac{\partial f}{\partial h_t}\right)^\top a_t$$

Step 4: Accumulate the parameter gradient:

$$\nabla_w L = \int_0^T \left(\frac{\partial f}{\partial w}\right)^\top a_t \, dt$$

For Neural SDEs, the authors instead use pathwise forward sensitivity, propagated alongside the SDE.

Training Neural SDE: Objective and Approach

Training Objective for the Neural SDE:

$$dh_t = f(h_t, t; w)\, dt + G(h_t, t; w)\, dB_t, \qquad h_{t_0} = h_0.$$

Since the hidden state depends on the Brownian path B, we minimize the expected terminal loss:

$$L(w) = \mathbb{E}_B[\ell(h_{t_1}(w, B)) \mid h_0], \qquad w^* = \arg\min_w L(w).$$

Gradient Computation

To train the network, we need $\nabla_w L(w)$. Under suitable regularity conditions, we can exchange differentiation and expectation:

$$\nabla_w L(w) = \mathbb{E}_B[\nabla_w \ell(h_{t_1}) \mid h_0]$$

Then, applying the chain rule:

$$\nabla_w \ell(h_{t_1}) = \left(\frac{\partial h_{t_1}}{\partial w}\right)^\top \nabla_h \ell(h_{t_1}).$$

Forward Sensitivity Approach: Instead of solving an adjoint equation backward, define the state sensitivity:

$$\beta_t = \frac{\partial h_t}{\partial w} \in \mathbb{R}^{n \times d}, \qquad w \in \mathbb{R}^d.$$

For M independent Brownian paths, approximate the expected gradient using Monte Carlo:

$$\widehat{\nabla_w L} = \frac{1}{M} \sum_{i=1}^{M} (\beta_{t_1}^{(i)})^\top \nabla_h \ell(h_{t_1}^{(i)}).$$

Derivation of the Sensitivity SDE

First, express the state dynamics in integral form:

$$h_t = h_0 + \int_{t_0}^{t} f(h_s, s; w)\, ds + \int_{t_0}^{t} G(h_s, s; w)\, dB_s.$$

To calculate the sensitivity $\beta_t = \partial h_t / \partial w$, apply the chain rule to both f and G:

$$\frac{\mathrm{d}}{\mathrm{d}w} f(h_t, t; w) = \underbrace{\frac{\partial}{\partial w} f(h_t, t; w)}_{f_w} + \underbrace{\frac{\partial}{\partial h_t} f(h_t, t; w)\, \beta_t}_{f_h},$$

$$\frac{\mathrm{d}}{\mathrm{d}w} G(h_t, t; w) = \underbrace{\frac{\partial}{\partial w} G(h_t, t; w)}_{G_w} + \underbrace{\frac{\partial}{\partial h_t} G(h_t, t; w)\, \beta_t}_{G_h}.$$

Here, f_w, G_w represent direct parameter dependence, while $f_h \beta_t, G_h \beta_t$ represent dependence through the hidden state.

Differentiating the integral equation gives

$$\beta_t = \beta_{t_0} + \int_{t_0}^{t} (f_w + f_h \beta_s)\, ds + \int_{t_0}^{t} (G_w + G_h \beta_s)\, dB_s.$$

Therefore, the sensitivity follows another SDE:

$$d\beta_t = (f_w + f_h \beta_t)\, dt + (G_w + G_h \beta_t)\, dB_t.$$

Joint Forward Integration and Gradient Computation

Solve the State and Sensitivity Together

Since both h_t and β_t satisfy SDEs, we can integrate them simultaneously:

$$\boxed{\begin{aligned} dh_t &= f(h_t, t; w)\, dt + G(h_t, t; w)\, dB_t, \\ d\beta_t &= (f_w + f_h \beta_t)\, dt + (G_w + G_h \beta_t)\, dB_t. \end{aligned}}$$

with initial conditions $h_{t_0} = h_0$ and $\beta_{t_0} = 0$.

Compute the Gradient at the Final Time

First, integrate forward using the same Brownian path:

$$(h_{t_0}, \beta_{t_0}) \xrightarrow{\text{SDE solver}} (h_{t_1}, \beta_{t_1}).$$

Then, use the terminal sensitivity and chain rule:

$$\nabla_w \ell(h_{t_1}) = (\beta_{t_1})^\top \nabla_h \ell(h_{t_1}).$$

Finally, average the gradients over M independent Brownian paths to estimate $\nabla_w L(w)$.

Method	Memory vs. solver steps N
Backpropagation through solver	$\mathcal{O}(N)$
Forward sensitivity SDE	$\mathcal{O}(1)$

Key Idea: Unlike the adjoint method, which propagates sensitivity backward, this approach propagates parameter sensitivity forward without storing all intermediate solver states.

Robustness Analysis

Robustness Analysis: Perturbation Dynamics

Goal: Analyze the stability of Neural SDE, showing that the randomness term can indeed improve the robustness of the model against small input perturbations.

Well-posedness assumptions

$$\|f(x,t)\| + \|G(x,t)\| \le c_1(1+\|x\|), \quad c_1 > 0 \quad (f \text{ and } G \text{ are at most linear})$$

$$\|f(x,t) - f(y,t)\| + \|G(x,t) - G(y,t)\| \le c_2\|x-y\|, \quad c_2 > 0 \quad (f \text{ and } G \text{ are } c_2\text{-Lipschitz})$$

These conditions ensure existence and uniqueness of the SDE solution.

Original and perturbed trajectories

$$\begin{aligned} \mathrm{d}h_t &= f(h_t,t)\,\mathrm{d}t + G(h_t,t)\,\mathrm{d}B_t, \\ \mathrm{d}h_t^e &= f(h_t^e,t;w)\,\mathrm{d}t + G(h_t^e,t;v)\,\mathrm{d}B_t', \\ h_0^e &= h_0 + \varepsilon_0, \quad \|\varepsilon_0\| \le \delta \end{aligned}$$

where B_t' is Brownian motions for the SDE associated with initialization h_0^e.

Perturbation SDE ($\varepsilon_t = h_t^e - h_t$)

$$\mathrm{d}\varepsilon_t = \big[f(h_t^e,t;w) - f(h_t,t)\big]\,\mathrm{d}t + \big[G(h_t^e,t;v) - G(h_t,t)\big]\,\mathrm{d}B_t = f_\Delta(\varepsilon_t,t)\,\mathrm{d}t + G_\Delta(\varepsilon_t,t)\,\mathrm{d}B_t.$$

The Brownian motions B_t and B_t' have the same sample path for both initialization $B_t = B_t'$.

Desired robustness property

$$\|\varepsilon_0\| \ll 1 \quad \Longrightarrow \quad \|\varepsilon_t\| \text{ remains small} \quad \Longrightarrow \quad \lim_{t\to\infty} \|\varepsilon_t\| = 0 \quad \text{a.s.}$$

Why Analyze the Zero-Error Solution?

Consider the perturbation dynamics:

$$d\varepsilon_t = f_\Delta(\varepsilon_t, t)\,dt + G_\Delta(\varepsilon_t, t)\,dB_t.$$

When $\varepsilon_t = 0$, the perturbed and original hidden states are identical. Therefore,

$$f_\Delta(0, t) = f(h_t, t) - f(h_t, t) = 0,$$
$$G_\Delta(0, t) = G(h_t, t) - G(h_t, t) = 0.$$

Thus, $\boxed{\varepsilon_t \equiv 0}$ is an equilibrium of the perturbation error, even though the hidden states may continue evolving.

However, Equilibrium Does Not Guarantee Stability

Consider a simple deterministic example:

$$d\varepsilon_t = a\varepsilon_t\,dt \quad \Longrightarrow \quad \varepsilon_t = \varepsilon_0 e^{at}.$$

$a < 0$	$\varepsilon_t \to 0$	Stable
$a > 0$	$\lvert\varepsilon_t\rvert \to \infty$	Unstable

In both cases, zero is an equilibrium, but a small perturbation may either decay or grow over time.

Stability Analysis for Neural SDE

Therefore, we need to determine under what conditions small initial perturbations decay:

$$\boxed{\varepsilon_0 \approx 0 \quad \Longrightarrow \quad \varepsilon_t \xrightarrow[t\to\infty]{} 0 \quad \text{a.s.}}$$

Stochastic Stability Analysis: Motivation and Assumptions

For multidimensional nonlinear SDEs, closed-form solutions are generally unavailable. Instead, we analyze asymptotic stability using stochastic Lyapunov theory.

$$d\varepsilon_t = f_\Delta(\varepsilon_t, t)\, dt + G_\Delta(\varepsilon_t, t)\, dB_t, \qquad \varepsilon_0 \neq 0.$$

Assume f_Δ and G_Δ are Borel measurable and satisfy:

$$\begin{aligned} \text{Linear growth:} \quad & \|f_\Delta(\varepsilon, t)\| + \|G_\Delta(\varepsilon, t)\| \leq c_1(1 + \|\varepsilon\|), \\ \text{Lipschitz:} \quad & \|f_\Delta(\varepsilon, t) - f_\Delta(\eta, t)\| + \|G_\Delta(\varepsilon, t) - G_\Delta(\eta, t)\| \leq c_2\|\varepsilon - \eta\|. \end{aligned}$$

These assumptions ensure existence and uniqueness of the perturbation SDE solution, allowing us to apply stochastic Lyapunov stability analysis.

Lemma A.1: Well-Posedness of the Perturbation SDE

Consider the original Lipschitz assumption: for all $x, y \in \mathbb{R}^n$ and $t \geq 0$:

$$\|f(x,t) - f(y,t)\| + \|G(x,t) - G(y,t)\| \leq c_2\|x - y\|.$$

Applying the Lipschitz assumption with $x = h_t + \varepsilon$ and $y = h_t$:

$$\begin{aligned}\|f_\Delta(\varepsilon,t)\| + \|G_\Delta(\varepsilon,t)\| &= \|f(h_t+\varepsilon,t) - f(h_t,t)\| + \|G(h_t+\varepsilon,t) - G(h_t,t)\| \\ &\leq c_2\|(h_t+\varepsilon) - h_t\| \leq c_2(1+\|\varepsilon\|).\end{aligned}$$

$$\boxed{\|f_\Delta(\varepsilon,t)\| + \|G_\Delta(\varepsilon,t)\| \leq c_2(1+\|\varepsilon\|).}$$

Consider two perturbations $\varepsilon, \widetilde{\varepsilon} \in \mathbb{R}^n$. From the definitions:

$$f_\Delta(\varepsilon,t) - f_\Delta(\widetilde{\varepsilon},t) = \big[f(h_t+\varepsilon,t) - f(h_t,t)\big] - \big[f(h_t+\widetilde{\varepsilon},t) - f(h_t,t)\big] = f(h_t+\varepsilon,t) - f(h_t+\widetilde{\varepsilon},t).$$

Similarly,

$$G_\Delta(\varepsilon,t) - G_\Delta(\widetilde{\varepsilon},t) = G(h_t+\varepsilon,t) - G(h_t+\widetilde{\varepsilon},t).$$

Apply the original Lipschitz condition

$$\|f_\Delta(\varepsilon,t) - f_\Delta(\widetilde{\varepsilon},t)\| + \|G_\Delta(\varepsilon,t) - G_\Delta(\widetilde{\varepsilon},t)\| \leq c_2\|(h_t+\varepsilon) - (h_t+\widetilde{\varepsilon})\| = c_2\|\varepsilon - \widetilde{\varepsilon}\|.$$

Conclusion of Lemma A.1

$$\begin{aligned}\text{Linear growth:} \quad & \|f_\Delta\| + \|G_\Delta\| \leq c_2(1+\|\varepsilon\|), \\ \text{Lipschitz:} \quad & \|\Delta f_\Delta\| + \|\Delta G_\Delta\| \leq c_2\|\varepsilon - \widetilde{\varepsilon}\|.\end{aligned}$$

Therefore, f_Δ and G_Δ satisfy the standard growth and Lipschitz conditions for an SDE.

Definition 3.1: Stability of the Zero-Error Solution

Consider the perturbation dynamics:

$$d\varepsilon_t = f_\Delta(\varepsilon_t, t)\, dt + G_\Delta(\varepsilon_t, t)\, dB_t.$$

Stochastic Stability: Perturbations Remain Small
For any $\alpha \in (0,1)$ and $r > 0$, there exists $\delta > 0$ such that

$$\boxed{\|\varepsilon_0\| \le \delta \implies \Pr\{\|\varepsilon_t\| < r,\ \forall t \ge 0\} \ge 1 - \alpha.}$$

Stochastic Asymptotic Stability: Perturbations Vanish
If, in addition, for any $\alpha \in (0,1)$, there exists $\delta > 0$ such that

$$\boxed{\|\varepsilon_0\| \le \delta \implies \Pr\left\{\lim_{t\to\infty} \|\varepsilon_t\| = 0\right\} \ge 1 - \alpha,}$$

Almost-Sure Exponential Stability: Exponential Decay
The stronger condition requires

$$\boxed{\limsup_{t\to\infty} \frac{1}{t} \log \|\varepsilon_t\| < 0 \quad \text{a.s.}, \qquad \forall \varepsilon_0 \neq 0.}$$

This means perturbations decay exponentially fast along almost every sample path.
Why Exponential Stability? Unlike asymptotic stability, exponential stability not only guarantees that perturbations vanish, but also characterizes how fast they decay.

Itô's Formula: Derivation for a General SDE

1. Consider a general vector SDE

$$dx_t = f(x_t, t)\,dt + G(x_t, t)\,dB_t, \qquad V = V(x_t, t),$$

where B_t is standard Brownian motion.

2. Apply the second-order Taylor expansion

$$dV = V_t\,dt + (\nabla V)^\top dx_t + \frac{1}{2}\,\mathrm{Tr}\left(\nabla^2 V\, dx_t dx_t^\top\right).$$

3. Use Itô multiplication rules

$$(dt)^2 = 0, \qquad dt\,dB_t^\top = 0, \qquad dB_t\,dB_t^\top = I\,dt.$$

Substituting $dx_t = f\,dt + G\,dB_t$:

$$dx_t dx_t^\top = (f\,dt + G\,dB_t)(f\,dt + G\,dB_t)^\top = G(dB_t\,dB_t^\top)G^\top = GG^\top\,dt.$$

4. Obtain Itô's formula

$$\boxed{dV = \underbrace{\left[V_t + (\nabla V)^\top f + \frac{1}{2}\,\mathrm{Tr}(G^\top \nabla^2 V G)\right]}_{\mathcal{L}V}\,dt + \underbrace{(\nabla V)^\top G\,dB_t}_{\text{Stochastic term}}.}$$

Key idea: Unlike the ordinary chain rule, Itô's formula includes a second-order correction due to Brownian noise.

Theorem 3.2: Almost-Sure Exponential Stability

Objective: Establish sufficient conditions under which the perturbation converges to zero exponentially, without explicitly solving the SDE.

$$d\varepsilon_t = f_\Delta(\varepsilon_t, t)\,dt + G_\Delta(\varepsilon_t, t)\,dB_t, \qquad \varepsilon_0 \neq 0.$$

Suppose there exists a smooth, positive-definite function $V(\varepsilon, t)$ and constants $p > 0$, $c_1 > 0$, $c_2 \in \mathbb{R}$, $c_3 \geq 0$ such that

(i) $V(\varepsilon, t) \geq c_1 \|\varepsilon\|^p$, Measures the perturbation magnitude; small V implies small error.

(ii) $\mathcal{L}V = V_t + V_\varepsilon f_\Delta + \frac{1}{2}\operatorname{Tr}(G_\Delta^\top V_{\varepsilon\varepsilon} G_\Delta) \leq c_2 V$, Bounds the growth of V due to drift and diffusion.

(iii) $\|V_\varepsilon G_\Delta\|^2 \geq c_3 V^2$, Bounds the stochastic variation of V from below.

Together: (i) relates stability to the error, (ii) bounds Lyapunov growth, and (iii) provides the negative Itô correction in $\log V$, yielding exponential stability if $c_3 > 2c_2$.

$$\boxed{\limsup_{t\to\infty} \frac{1}{t} \log \|\varepsilon_t\| \leq -\frac{c_3 - 2c_2}{2p} \quad \text{a.s.}}$$

$c_3 > 2c_2 \quad \Longrightarrow \quad$ Almost-sure exponential stability.

Proof of Theorem 3.2: Deriving Eq. (21)

Connection between V and exponential stability:

$$V(\varepsilon, t) \geq c_1 \|\varepsilon\|^p, \qquad c_1 > 0, \quad p > 0.$$

Then, for $\varepsilon_t \neq 0$:

$$\log \|\varepsilon_t\| \leq \frac{1}{p} \left[\log V(\varepsilon_t, t) - \log c_1\right].$$

Thus, controlling the long-term growth of $\log V$ provides a bound on the exponential growth of $\|\varepsilon_t\|$.

Preliminary Result (Lemma A.2):

Before applying Itô's formula to $\log V$, we must ensure that $V(\varepsilon_t, t) > 0$, since $\log(0)$ is undefined.

Lemma A.2 guarantees that if the initial perturbation is nonzero, the error never reaches exactly zero at any finite time, almost surely:

$$\boxed{\varepsilon_0 \neq 0 \quad \Longrightarrow \quad \Pr\{\varepsilon_t \neq 0, \ \forall t \geq 0\} = 1.}$$

Since $V(\varepsilon, t) > 0$, this guarantees that $\log V(\varepsilon_t, t)$ is well-defined, allowing us to apply Itô's formula.

Apply Itô's formula to $\log V$

$$dV = \mathcal{L}V \, dt + V_\varepsilon G_\Delta \, dB_t, \qquad (dV)^2 = \|V_\varepsilon G_\Delta\|^2 dt,$$

$$d \log V = \frac{1}{V} dV - \frac{1}{2V^2} (dV)^2 = \left[\frac{\mathcal{L}V}{V} - \frac{\|V_\varepsilon G_\Delta\|^2}{2V^2}\right] dt + \frac{V_\varepsilon G_\Delta}{V} \, dB_t.$$

Proof of Theorem 3.2: Deriving Eq. (21)

Integrate over $[0, t]$

$$\log V(\varepsilon_t, t) = \log V(\varepsilon_0, 0) + \int_0^t \frac{\mathcal{L}V}{V}\, ds - \frac{1}{2}\int_0^t \frac{\|V_\varepsilon G_\Delta\|^2}{V^2}\, ds + M(t), \quad M(t) = \int_0^t \frac{V_\varepsilon G_\Delta}{V}\, dB_s.$$

Apply the Lyapunov condition

$$\mathcal{L}V \le c_2 V \quad \Longrightarrow \quad \int_0^t \frac{\mathcal{L}V}{V}\, ds \le c_2 t.$$

$$\log V(\varepsilon_t, t) \le \log V(\varepsilon_0, 0) + c_2 t + M(t) - \frac{1}{2}\int_0^t \frac{\|V_\varepsilon G_\Delta\|^2}{V^2}\, ds. \tag{21}$$

$$\log V(\varepsilon_t, t) \le \log V(\varepsilon_0, 0) + c_2 t + M(t) - \frac{1}{2}\langle M\rangle_t, \quad \langle M\rangle_t = \int_0^t \frac{\|V_\varepsilon G_\Delta\|^2}{V^2}\, ds$$

$M(t)$ is the stochastic term, and $\langle M\rangle_t$ is its quadratic variation.

Proof of Theorem 3.2: Controlling the Stochastic Term

Apply the Exponential-Martingale Inequality: For any $\alpha > 0$ and $c > 0$, the exponential-martingale inequality gives

$$\mathbb{P}\left\{ \sup_{0 \le t \le n} \left[M(t) - \frac{\alpha}{2} \langle M \rangle_t \right] > c \right\} \le e^{-\alpha c}$$

To apply the Borel–Cantelli lemma, choose c such that the probability bound is summable:

$$e^{-\alpha c} = \frac{1}{n^2} \quad \Longrightarrow \quad c = \frac{2}{\alpha} \log n$$

$$\mathbb{P}\left\{ \sup_{0 \le t \le n} \left[M(t) - \frac{\alpha}{2} \langle M \rangle_t \right] > \frac{2}{\alpha} \log n \right\} \le \frac{1}{n^2}. \tag{22}$$

Thus, the probability of exceeding this bound decreases as n increases.

Apply the Borel–Cantelli Lemma:

Since $\sum_{n=2}^{\infty} 1/n^2 < \infty$, the Borel–Cantelli lemma implies that exceeding the bound in Eq. (22) can occur only finitely many times, almost surely. Therefore, for sufficiently large n, the following bound holds for all $0 \le t \le n$, almost surely:

$$M(t) \le \frac{2}{\alpha} \log n + \frac{\alpha}{2} \langle M \rangle_t, \quad 0 \le t \le n. \tag{23}$$

Proof of Theorem 3.2: Exponential Stability

Combining Eqs. (21) and (23):

$$\log V(\varepsilon_t, t) \leq \log V(\varepsilon_0, 0) + c_2 t + \frac{2}{\alpha} \log n - \frac{1-\alpha}{2} \langle M \rangle_t.$$

Using the third condition of Theorem 3.2,

$$\|V_\varepsilon G_\Delta\|^2 \geq c_3 V^2 \quad \Longrightarrow \quad \langle M \rangle_t \geq c_3 t.$$

$$\log V(\varepsilon_t, t) \leq \log V(\varepsilon_0, 0) - \frac{(1-\alpha)c_3 - 2c_2}{2} t + \frac{2}{\alpha} \log n \tag{24}$$

Divide both sides by t. For $n-1 \leq t \leq n$ and sufficiently large $n \geq n_0(\omega)$, we obtain:

$$\frac{1}{t} \log V(\varepsilon_t, t) \leq -\frac{(1-\alpha)c_3 - 2c_2}{2} + \frac{\log V(\varepsilon_0, 0) + \frac{2}{\alpha} \log n}{n-1}. \tag{25}$$

which consequently implies

$$\limsup_{t \to \infty} \frac{1}{t} \log V(\varepsilon_t, t) \leq -\frac{(1-\alpha)c_3 - 2c_2}{2} \quad \text{a.s.} \tag{26}$$

Since $\alpha \in (0,1)$ is arbitrary, let $\alpha \to 0$. Using condition (1):

$$c_1 \|\varepsilon_t\|^p \leq V(\varepsilon_t, t) \quad \Longrightarrow \quad p \log \|\varepsilon_t\| \leq \log V(\varepsilon_t, t) - \log c_1.$$

$$\limsup_{t \to \infty} \frac{1}{t} \log \|\varepsilon_t\| \leq -\frac{c_3 - 2c_2}{2p} \quad \text{a.s.} \tag{15}$$

Hence, if $c_3 > 2c_2$, the perturbation converges to zero exponentially almost surely.

Multiplicative Noise: Perturbation Dynamics

Consider two trajectories driven by the same noise

For multiplicative diffusion $G(h,t) = \sigma h$, with $\sigma > 0$ and scalar Brownian motion:

$$dh_t = f(h_t, t)\, dt + \sigma h_t\, dB_t,$$
$$d\widetilde{h}_t = f(\widetilde{h}_t, t)\, dt + \sigma \widetilde{h}_t\, dB_t.$$

Derive the perturbation dynamics

Define $\varepsilon_t = \widetilde{h}_t - h_t$. Subtracting:

$$d\varepsilon_t = [f(\widetilde{h}_t, t) - f(h_t, t)]\, dt + \sigma(\widetilde{h}_t - h_t)\, dB_t = f_\Delta(\varepsilon_t, t)\, dt + \sigma\varepsilon_t\, dB_t,$$

$$f_\Delta(\varepsilon_t, t) = f(h_t + \varepsilon_t, t) - f(h_t, t).$$

Apply the Lipschitz assumption

Assume the drift f is L-Lipschitz:

$$\|f(x,t) - f(y,t)\| \le L\|x - y\|.$$

Setting $x = h_t + \varepsilon_t$ and $y = h_t$ gives

$$\boxed{\|f_\Delta(\varepsilon_t, t)\| \le L\|\varepsilon_t\|.}$$

Goal: Determine when ε_t converges exponentially to zero despite drift-induced growth.

Multiplicative Noise: Verify Lyapunov Conditions

Apply Theorem 3.2 to

$$d\varepsilon_t = f_\Delta(\varepsilon_t, t)\,dt + \sigma\varepsilon_t\,dB_t$$

Choose $V(\varepsilon, t) = \|\varepsilon\|^2$, giving

$$V_\varepsilon = 2\varepsilon^\top, \qquad V_{\varepsilon\varepsilon} = 2I, \qquad V_t = 0.$$

Condition (i): Lower bound on V

$$V(\varepsilon, t) = \|\varepsilon\|^2 \geq c_1\|\varepsilon\|^p \quad \Rightarrow \quad \boxed{c_1 = 1, \quad p = 2.}$$

Condition (ii): Bound the Lyapunov generator $\mathcal{L}V \leq c_2 V$

$$\mathcal{L}V = V_\varepsilon f_\Delta + \frac{1}{2}G_\Delta^\top V_{\varepsilon\varepsilon} G_\Delta = 2\varepsilon^\top f_\Delta + \sigma^2\|\varepsilon\|^2 \leq 2\|\varepsilon\|\|f_\Delta\| + \sigma^2\|\varepsilon\|^2 \leq (2L + \sigma^2)V.$$

$$\boxed{c_2 = 2L + \sigma^2}$$

Condition (iii): Bound the diffusion contribution $\|V_\varepsilon G_\Delta\|^2 \geq c_3 V^2$

$$\|V_\varepsilon(\varepsilon, t)G_\Delta(\varepsilon, t)\|^2 = \left|2\varepsilon^\top(\sigma\varepsilon)\right|^2 = 4\sigma^2\|\varepsilon\|^4 = 4\sigma^2 V(\varepsilon, t)^2$$

$$\boxed{c_3 = 4\sigma^2}$$

Corollary 3.2.1: Exponential Stability

1. Recall the result of Theorem 3.2

$$\boxed{\limsup_{t\to\infty} \frac{1}{t} \log \|\varepsilon_t\| \leq -\frac{c_3 - 2c_2}{2p} \quad \text{a.s.}}$$

2. Substitute the constants

$$-\frac{c_3 - 2c_2}{2p} = -\frac{4\sigma^2 - 2(2L + \sigma^2)}{4} = -\frac{2\sigma^2 - 4L}{4} = L - \frac{\sigma^2}{2}$$

$$\boxed{\limsup_{t\to\infty} \frac{1}{t} \log \|\varepsilon_t\| \leq L - \frac{\sigma^2}{2} \quad \text{a.s.}}$$

3. Sufficient condition for stability

For exponential decay, the upper bound must be negative:

$$L - \frac{\sigma^2}{2} < 0 \quad \Longleftrightarrow \quad \boxed{\sigma^2 > 2L.}$$

Interpretation

Without noise ($\sigma = 0$), the bound does not guarantee decay. Sufficiently strong multiplicative noise ($\sigma^2 > 2L$) guarantees that perturbations decay exponentially almost surely.

Experiments

Experimental setup and diffusion choices

Objective: Evaluate whether noise injection improves generalization and robustness over Neural ODE.

Model	Diffusion function $G(h_t, t)$
Neural ODE	0
Additive Neural SDE	$\sigma_t I$
Multiplicative Neural SDE	$\sigma_t h_t$
Dropout-style Neural SDE	$\sigma_t \operatorname{diag}\{f(h_t, t; w)\}$

- Select an appropriate noise intensity σ_t for each diffusion and evaluate accuracy with and without TTN.
- **TTN = Testing Time Noise:** keep the SDE diffusion active at test time, run the same input through several Brownian sample paths, and average the predictions.

Data	Model size	Accuracy@1 — w/o TTN				Accuracy@1 — w/ TTN			
		ODE	Additive	Multiplicative	Dropout	ODE	Additive	Multiplicative	Dropout
CIFAR-10	115 K	81.63	83.65	83.26	83.60	–	83.89	83.76	**84.55**
STL-10	2.44 M	58.03	61.23	60.54	61.26	–	62.11	**62.58**	62.13
Tiny-ImageNet	2.49 M	45.19	45.25	46.94	47.04	–	45.39	46.65	**47.81**

- Neural SDE outperforms Neural ODE on all three datasets.
- Stochasticity acts as a regularizer, helping reduce overfitting and improve generalization.

Experiment 2: Non-adversarial robustness

Objective: Evaluate robustness against common image corruptions unseen during training.

Experimental procedure:

- Train models using only clean images.
- Test using CIFAR-10-C and Tiny-ImageNet-C.
- Apply corruptions such as Gaussian noise, motion blur, and fog.
- Evaluate five severity levels, from mild (Level 1) to severe (Level 5).

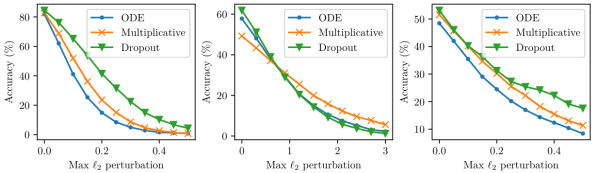
Data	Noise type	mild corrupt ← Accuracy → severe corrupt				
		Level 1	Level 2	Level 3	Level 4	Level 5
CIFAR10-C†	ODE	75.89	70.59	66.52	60.91	53.02
	Dropout	77.02	71.58	67.21	61.61	53.81
	Dropout+TTN	**79.07**	**73.98**	**69.74**	**64.19**	**55.99**
TinyImageNet-C†	ODE	23.01	19.18	15.20	**12.20**	**9.88**
	Dropout	22.85	18.94	14.64	11.54	9.09
	Dropout+TTN	**23.84**	**19.89**	**15.28**	12.08	9.44

- Dropout-style SDE with TTN outperforms ODE in 8 of the 10 reported cases.
- Improvements are not universal, especially under severe Tiny-ImageNet corruptions.

Experiment 3: Adversarial Robustness

Objective: Evaluate whether Neural SDE improves robustness against designed input perturbations.
Attack: Multi-step Projected Gradient Descent (PGD)

- Introduces small changes to input images to maximize classification errors.
- Restricts the perturbation magnitude to keep the attacked images close to the originals.
- Increasing the perturbation limit produces stronger attacks.



CIFAR-10 (left), STL-10 (middle), Tiny-ImageNet (right)

Observed Results:

- Accuracy decreases as attack strength increases.
- Neural SDE generally outperforms Neural ODE.
- Dropout-style noise shows the strongest overall robustness, particularly on CIFAR-10 and Tiny-ImageNet.

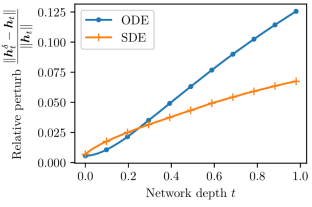
Stochastic noise injection improves resistance to adversarial attacks, helping maintain higher classification accuracy under input perturbations.

Experiment 4: Hidden-State Perturbations

Objective: Investigate how adversarial input perturbations propagate through the hidden states of Neural ODE and Neural SDE.

Experimental procedure (CIFAR-10):

1. Select an original image x and record its hidden states h_t at different network depths.
2. Generate an adversarial image $x_{\mathrm{adv}} = x + \delta_x$ using PGD.
3. Record the perturbed hidden states $\widetilde{h}_t$ at the same depths.
4. Measure the relative hidden-state perturbation: $\Longrightarrow \quad R(t) = \frac{\|\widetilde{h}_t - h_t\|_2}{\|h_t\|_2}$



Observed results:

- Neural ODE: perturbations grow rapidly with network depth.
- Neural SDE: dropout-style diffusion slows perturbation growth.
- Final relative perturbation is approximately 48% lower for Neural SDE.

Neural SDE reduces hidden-state sensitivity to adversarial inputs, providing empirical support for its improved robustness.

Conclusion

Proposed Method

- Extend Neural ODEs to Neural SDEs by introducing stochastic noise into continuous-depth networks.
- Incorporate regularization methods such as dropout and Gaussian noise injection.

Theoretical Contributions

- Develop an efficient forward sensitivity method for training Neural SDEs.
- Establish conditions for almost-sure exponential stability using Lyapunov theory.

Experimental Findings

- Neural SDE generally improves generalization and robustness over the tested Neural ODE baseline.
- Improvements depend on the diffusion structure and noise intensity.

Final Takeaway:
Properly designed stochastic noise can stabilize perturbation dynamics, but stability does not automatically guarantee better classification accuracy.

Thank You!

Questions?

Appendix
Lemma A.2

Lemma A.2: Non-Attainment of Zero

Before analyzing exponential stability, we first show that a nonzero perturbation cannot reach exactly zero in finite time, almost surely:

$$\boxed{\Pr\{\varepsilon_t \neq 0,\ \forall t \geq 0\} = 1, \qquad \varepsilon_0 \neq 0.}$$

This allows us to work with $\log \|\varepsilon_t\|$ when proving almost-sure exponential stability.

Proof by contradiction

Define the first hitting time of zero:

$$\tau = \inf\{t \geq 0 : \varepsilon_t = 0\}.$$

Suppose the error reaches zero in finite time with positive probability:

$$\mathbb{P}(\tau < \infty) > 0.$$

Then, for some finite $T > 0$ and sufficiently large $\theta > 1$ (bounds the error trajectory before reaching zero), define

$$A = \left\{ \tau < T,\ \sup_{0 \leq s \leq \tau} \|\varepsilon_s\| \leq \theta - 1 \right\}, \qquad \mathbb{P}(A) > 0.$$

Goal: Show $\mathbb{P}(A) = 0$, contradicting $\mathbb{P}(A) > 0$.

Bound the perturbation coefficients

By Assumption 2, there exists $K_\theta > 0$ such that

$$\|f_\Delta(\varepsilon, t)\| + \|G_\Delta(\varepsilon, t)\| \leq K_\theta \|\varepsilon\|, \tag{17}$$

for $\|\varepsilon\| \leq \theta$ and $0 \leq t \leq T$.

Lemma A.2: Singular Lyapunov Function

Choose a Function That Detects Approaching Zero

$$V(\varepsilon) = (\varepsilon^\top \varepsilon)^{-1/2} = \|\varepsilon\|^{-1} \quad \xrightarrow{\|\varepsilon\| \to 0} \quad \infty.$$

$$V_\varepsilon = -\frac{1}{2}(\varepsilon^\top \varepsilon)^{-3/2}(2\varepsilon^\top) = -\|\varepsilon\|^{-3}\varepsilon^\top,$$

$$V_{\varepsilon\varepsilon} = -\|\varepsilon\|^{-3} I + 3\|\varepsilon\|^{-5}\varepsilon\varepsilon^\top.$$

$$\mathcal{L}V = V_\varepsilon f_\Delta + \frac{1}{2}\operatorname{Tr}(G_\Delta^\top V_{\varepsilon\varepsilon} G_\Delta).$$

Using $\operatorname{Tr}(G_\Delta^\top G_\Delta) = \|G_\Delta\|^2$ and $\operatorname{Tr}(G_\Delta^\top \varepsilon\varepsilon^\top G_\Delta) = \|\varepsilon^\top G_\Delta\|^2$:

$$\mathcal{L}V = -\|\varepsilon\|^{-3}\varepsilon^\top f_\Delta + \frac{1}{2}\left[-\|\varepsilon\|^{-3}\|G_\Delta\|^2 + 3\|\varepsilon\|^{-5}\|\varepsilon^\top G_\Delta\|^2\right].$$

Bound the Generator Using Eq. (17):

By Cauchy–Schwarz:

$$|\varepsilon^\top f_\Delta| \le \|\varepsilon\|\|f_\Delta\|, \qquad \|\varepsilon^\top G_\Delta\|^2 \le \|\varepsilon\|^2\|G_\Delta\|^2.$$

Therefore,

$$\mathcal{L}V \le \|\varepsilon\|^{-2}\|f_\Delta\| + \|\varepsilon\|^{-3}\|G_\Delta\|^2 \le K_\theta\|\varepsilon\|^{-1} + K_\theta^2\|\varepsilon\|^{-1}.$$

Since $V(\varepsilon) = \|\varepsilon\|^{-1}$,

$$\boxed{\mathcal{L}V \le K_\theta(1 + K_\theta)V} \tag{18}$$

Lemma A.2: Stopping Time and Itô's Formula

Since $V(\varepsilon) = 1/\|\varepsilon\|$ is undefined at zero, we first restrict the process to $\delta \leq \|\varepsilon_t\| \leq \theta$.
For $0 < \delta < \|\varepsilon_0\| < \theta$, define

$$\boxed{\begin{aligned}\tau_\delta &= \inf\{t \geq 0 : \|\varepsilon_t\| \notin (\delta, \theta)\},\\ \nu_\delta &= \min\{\tau_\delta, T\}.\end{aligned}}$$

Here, τ_δ is the first exit time from (δ, θ), and ν_δ also limits the process to the finite interval $[0, T]$.

2. Introduce a Discounted Function
From Eq. (18), $\mathcal{L}V \leq aV$, where $a = K_\theta(1 + K_\theta)$.
To eliminate this positive growth bound, define

$$\boxed{Y_t = e^{-at}V(\varepsilon_t).}$$

Applying Itô's product rule:

$$dY_t = e^{-at}(\mathcal{L}V - aV)\,dt + e^{-at}V_\varepsilon G_\Delta\,dB_t.$$

Since $\mathcal{L}V - aV \leq 0$, the drift of Y_t is nonpositive, so its expected value cannot increase over time.
Take Expectations Up to the Stopping Time ν_δ
Since the Brownian increments have zero mean, the stopped Itô integral has zero expectation. Also, $\mathcal{L}V - aV \leq 0$. Therefore,

$$\mathbb{E}[Y_{\nu_\delta}] = V(\varepsilon_0, 0) + \mathbb{E}\int_0^{\nu_\delta} e^{-as}(\mathcal{L}V - aV)\,ds \leq V(\varepsilon_0, 0) = \|\varepsilon_0\|^{-1}. \tag{19}$$

Lemma A.2: Final Contradiction

Recall that on A, the error reaches zero before time T without exceeding the bound $\theta - 1$:

$$A = \left\{ \tau < T,\ \sup_{0 \le s \le \tau} \|\varepsilon_s\| \le \theta - 1 \right\}, \qquad \Pr(A) > 0.$$

Since the trajectories are continuous, the error must first cross the level $\delta > 0$:

$$\nu_\delta = \tau_\delta < T, \quad \|\varepsilon_{\nu_\delta}\| = \delta \quad \Longrightarrow \quad V(\varepsilon_{\nu_\delta}) = \|\varepsilon_{\nu_\delta}\|^{-1} = \delta^{-1}.$$

From Eq. (19),

$$\mathbb{E}[Y_{\nu_\delta}] \le \|\varepsilon_0\|^{-1}$$

However, on the event A,

$$Y_{\nu_\delta} = e^{-a\nu_\delta}\delta^{-1} \ge e^{-aT}\delta^{-1}$$

Thus,

$$\|\varepsilon_0\|^{-1} \ge \mathbb{E}[Y_{\nu_\delta}] \ge \mathbb{E}[\mathbf{1}_A Y_{\nu_\delta}] \ge e^{-aT}\delta^{-1}\Pr(A)$$

Rearranging gives

$$\boxed{\Pr(A) \le \delta \|\varepsilon_0\|^{-1} e^{aT}} \tag{20}$$

$$\delta \to 0 \quad \Longrightarrow \quad \boxed{\Pr(A) = 0}$$

This contradicts $\Pr(A) > 0$. Therefore,

$$\Pr\{\varepsilon_t \ne 0,\ \forall t \ge 0\} = 1, \qquad \varepsilon_0 \ne 0.$$